SkillBridge

USER MANUAL

An AI-Powered Skill Intelligence and Placement Platform

A practical guide for Students, Faculty / T&P Officers,
Recruiters and Platform Administrators

Project Based Learning • Group 11
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Prepared from the supplied SkillBridge project report and source-code archive.

About This Manual

This manual explains the user-facing workflows available in the supplied SkillBridge application. It is intended for people using the application, not for developers maintaining its source code. Screen names in this guide follow the supplied frontend routes and components.

Important scope note
The supplied project report describes the intended platform and its five major modules. The uploaded ZIP contains the implementation snapshot inspected for this manual. Some pages use sample or mock data, and some external integrations depend on configuration. This guide identifies those limits rather than presenting planned capabilities as guaranteed live services.

Who should use it?

User type	Main tasks described in this manual
Student	Create an account, maintain a profile, add evidence, inspect skills, review skill gaps, use learning resources and explore role readiness.
Faculty / T&P Officer	Review cohort-level analytics, filter student readiness information and inspect common skill gaps.
Recruiter / Company	Select a job, review consented candidate matches, compare matched and missing skills, and update shortlist status.
Platform Administrator	Review platform metrics, manage users, inspect integrations, view audit records and adjust available settings.

How to read the instructions

- Follow numbered steps in order. Button and page names are shown in bold.
- A page may look different if the application is updated or configured for another environment.
- If a feature is unavailable, contact the person who operates your SkillBridge installation.

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1. Getting Started

SkillBridge brings student skill information and placement-preparation activities into one web application. The project report describes a workflow of Collect, Analyze, Improve and Match.

Stage	What it means to the user
Collect	Maintain a student profile and provide supporting information such as projects, coding profiles, certificates and other evidence.
Analyze	Review the skills identified from available information and the associated skill scores or verification labels.
Improve	Use skill-gap information and learning resources to decide what to study next.
Match	Compare the current skill profile with the requirements associated with a career role or job.

1.1 Accessing the application

- Open the SkillBridge web address supplied by your college, project team or system administrator.
- Use a modern browser such as Chrome, Edge or Firefox. Keep an internet connection available if the installation uses hosted services or external resources.
- The sign-in screen is the normal entry point. New student users can choose the registration link.

No public production URL was supplied
The ZIP is a source-code archive, not a hosted website address. This manual therefore does not invent a live URL. Ask the project administrator for the correct address for your installation.

1.2 Main navigation

The student workspace provides pages for Dashboard, Profile, Skills, Evidence, Skill Gap, Recommendations / Learning, Roles and Progress / Readiness. The available pages depend on the account role and the routes enabled in the installed build.

2. Account Registration and Login

2.1 Create a student account

The registration screen is labelled Student Registration. It asks for a full name, college email address, password and password confirmation.

- Open the application sign-in page and select the registration link (usually labelled **Create an account** or **Sign Up**).
- Enter your full name.
- Enter a valid college email address.
- Create a password with at least six characters.
- Enter the same password in the confirmation field.
- Select **Create Student Account**. If registration succeeds, the application navigates to the student dashboard.

2.2 Sign in

- Open the **Login** page.
- Enter the email address registered with SkillBridge.
- Enter your password.
- Select the sign-in button. After successful authentication, the application opens the permitted workspace.
- If the credentials are rejected, check the email and password and try again.

2.3 Demo access

The inspected login component contains a quick demo access action and prefilled sample student credentials. Availability depends on the running environment and its seeded database. Use demo access only when the project team has explicitly enabled it for demonstration. Do not reuse sample credentials for a real account.

Account safety
Never share your password. Use your own account when entering personal academic or contact information. If you suspect that someone else has accessed your account, contact the system administrator.

3. Student Workspace

After sign-in, the student dashboard provides a consolidated view of the student's profile summary, skill information, role-readiness information, recommended roles, learning summary and recent activity. The exact cards shown depend on available records and the selected data mode.

3.1 Understand the dashboard

Dashboard area	How to use it
Welcome / profile summary	Check that the displayed name and profile summary belong to you.
Skills overview	Review the skills displayed and use category filters where available.
Role readiness	Review the readiness information associated with a target role. Treat it as guidance, not a placement guarantee.
Recommended roles	Explore roles suggested by the available profile and skill information.
Learning summary	Review learning activity and progress where learning records are available.
Recent activity	Check recent changes or evidence-related activity displayed by the application.

3.2 Data mode

The dashboard source includes a data-mode selector with **Realtime** and **Sample** options. Realtime mode requests data through the configured application services. Sample mode is intended for demonstration and may display illustrative records. A sample-data action is also present in the source. Only use data-seeding controls when instructed by the administrator; do not assume sample records are your personal records.

> **Check before relying on a score**
> Confirm whether the dashboard is showing sample or realtime data. Scores and examples in sample mode should not be treated as verified personal results.

3.3 Moving between pages

Use the navigation menu/sidebar to open the relevant student page. If a page does not load, return to the dashboard and confirm that you are signed in with a role allowed to access that page.

4. Profile Management

The Profile page lets a student maintain personal, academic, career and social-profile details. The inspected form includes:

Profile group	Examples of fields
Personal / professional	Full name and email are displayed; headline, biography and contact number are available in the form.
Career preferences	Target job role and technical interests.
Academic information	College, department, degree, current year, graduation year and CGPA.
External profiles	GitHub, LinkedIn, LeetCode and portfolio links.

4.1 Update your profile

- Open **Profile** from the student navigation.
- Review each field carefully. Replace any sample or placeholder information with your own correct details.
- Enter technical interests as a comma-separated list if the field is presented as one text box.
- Check that external profile URLs point to your own accounts.
- Select **Save** or the save-profile action.
- Wait for the success or error message. If saving fails, check required fields and try again.

Important profile check
The source snapshot contains fallback example values in the profile form. Verify every prefilled value before saving. Do not leave example contact details, academic details or profile URLs in your account.

4.2 Keep information current

Update your target role when your career goal changes. Keep academic details and external profile links accurate so that later skill analysis is based on relevant information.

5. Skills and Evidence

5.1 View your skills

Open **Skills** (labelled Skill Competency in the page). The page displays evaluated skills, categories, scores, proficiency levels and evidence-related information where available. Use the search box to find a skill, choose the All / hard / soft filter tabs, and change the sorting option to review the list.

A proficiency label may include Beginner, Intermediate, Advanced or Provisional. Verification descriptions distinguish independently verified information, evidence-supported information and self-reported claims.

5.2 Add skill evidence

Evidence is supporting material for a skill or achievement. The evidence page includes an **Add Evidence** action, search and type filters, verification labels, and actions for viewing or downloading available files and deleting a record.

- Open **Evidence** from the student navigation, or select **Add Skill Evidence** from the Skills page.
- Select **Add Evidence** to open the evidence form.
- Choose the evidence type offered by the form (for example, project, certificate, internship or coding profile).
- Enter the requested title, description, URL or supporting details. Attach a file if the form asks for one.
- Review the information and submit the form.
- Return to the evidence list and check that the record appears. Review its verification label.

5.3 Search, filter and manage evidence

- Use the search field to locate an evidence record by its available text.
- Use the type filter to narrow the list.
- A Verified label means the record has a verification status in the application; Pending means verification is not complete; Unverified means it is a self-reported claim.
- Use the download action only when a downloadable file is attached.
- Delete only records you no longer need. The application asks for confirmation before deletion.

Evidence quality
Submit genuine, relevant and readable evidence. A URL or uploaded document is supporting information; do not describe it as independently verified unless the application explicitly shows that status.

6. Skill-Gap Analysis and Learning

6.1 Run a skill-gap analysis

The Skill Gap page loads available career-role definitions and runs an analysis for the selected role. It compares the skills represented in the student's available profile/evidence with the selected role's requirements.

- Open **Skill Gap** from the student navigation.
- Wait for the list of available roles to load.
- Select the career role you want to work toward.
- Allow the analysis to finish.
- Review each skill's status and the explanation shown. A status such as Meets Requirement, Partial or Gap indicates how the available skill information compares with the role requirement.
- Use the results to decide which skills need more evidence or further development.

The page source identifies a FastAPI AI service / deterministic rubric as the analysis source label. The exact analysis result depends on the configured services and the records available for the account.

6.2 Explore learning recommendations

Open **Recommendations** or **Learning**. The page provides a learning-resource list with filters for skill focus, difficulty and search text. It also supports starting a resource and updating progress.

- Open **Recommendations / Learning**.
- Choose a focus skill or keep the all-skills option.
- Choose a difficulty level if you want to narrow the list.
- Enter a keyword and submit the search if needed.
- Open a resource to review its details and provider information.
- Use the start / progress action available for that resource. Update progress as you complete the learning activity.

Learning links and providers
Some learning resources may open external websites. Review the destination before leaving SkillBridge. Completion tracking in the application records the progress you submit; it does not independently prove that an external course was completed.

7. Role Matching and Progress

7.1 Explore job roles

The **Roles / Job Role Matching** page presents role options and compares skills with role requirements. The inspected page contains example role data and a local analysis interaction; treat displayed sample match percentages as illustrative unless your installed build clearly indicates that live account data is being used.

- Choose a target role from the available role selector or role list.
- Review matched, partial and gap skills.
- Read the skill comparison rather than relying only on the overall percentage.
- Use the gap information to return to Skills, Evidence or Learning.

7.2 Review progress / readiness

The **Progress / Overall Readiness** page provides a summary-style view of skill development against a selected role. It displays skill rows with your score, required score and a status such as Met, Partial or Gap. The current source includes hard-coded illustrative skill rows and a short local analysis interaction; do not treat these values as a live verified assessment without confirmation from the administrator.

7.3 Understand scores

Term	Meaning
Skill score	A numeric representation of the skill information available to the platform. It is not a guarantee of expertise.
Required score	The comparison threshold associated with the selected role in the displayed analysis.
Matched / Met	The available score meets the displayed requirement.
Partial	The available score is close to, but does not fully meet, the displayed requirement.
Gap	The available score is below the displayed requirement or relevant evidence is missing.
Confidence / verification	An indication of how strongly the platform's result is supported by available evidence and checks.

No placement guarantee
Role matching and readiness are decision-support features. They do not guarantee an interview, job offer or placement. Final hiring decisions remain with the employer.

8. Faculty / T&P Workspace

The Faculty / T&P route is restricted to accounts with the faculty or administrator role. The supplied FacultyDashboard component is a cohort analytics view.

8.1 Review batch analytics

- Sign in with an authorized Faculty / T&P account.
- Open the Faculty / T&P analytics workspace.
- Use the Department filter to select a department or All Departments.
- Use the Batch filter to select a batch or All Batches.
- Use the Target Role filter to focus on a placement role.
- Select **Refresh Analytics** if you want to reload the current filter results.
- Review the total-student, placement-ready, average-readiness and needs-upskilling summary values.
- Inspect the batch-level skill distribution and common skill gaps table.

8.2 Interpret analytics

The dashboard displays aggregate information such as average skill scores, proficiency bands, readiness summaries and common gaps against a benchmark. These are cohort-level indicators; use them to guide training and student support, not as the sole basis for high-stakes decisions.

> **Access and data availability**
> If analytics fail to load, confirm that the account has the correct role and that the backend/API and database are available. The dashboard requests data from the configured Faculty API; it may show an error when the service is unavailable.

9. Recruiter Workspace

The Recruiter route is restricted to recruiter or administrator accounts. The supplied recruiter dashboard displays job requisitions and candidate matching information.

9.1 Review candidates for a job

- Sign in with an authorized recruiter account.
- Open the Recruiter workspace.
- Choose a job requisition from **Select Job Requisition**.
- Wait while the candidate matching information loads.
- Review each candidate's match score, confidence label, department, batch and target role.
- Compare the matched requirements and listed shortfalls.
- Read the explanation for the match before taking any action.

9.2 Update shortlist status

For a candidate, the dashboard provides actions labelled **Shortlist**, **Schedule Interview** and **Pass**. Choose the action that reflects the recruiter's decision. The dashboard then attempts to save the status and refresh the job's candidate list.

> **Candidate privacy**
> The recruiter interface indicates that candidates without consent are excluded. Use candidate information only for the recruitment purpose for which access was granted. Follow your institution's and organisation's data-handling rules.

9.3 If no candidates appear

Check that a job requisition is selected, the job has defined requirements, and eligible candidates have consented to be considered. If the page displays an error or remains empty, contact the administrator.

10. Administrator Workspace

The Admin route is restricted to administrator accounts. The supplied AdminPage provides the following tabs:

Admin area	Purpose
Dashboard	View platform-level summary information.
User Management	Review and manage user accounts using the available controls.
Integrations	Inspect the status of configured integration services.
Audit Trail	Review recorded platform events and activity.
Settings	Review or update available platform settings.

10.1 Manage the platform safely

- Use an administrator account only for platform administration.
- Confirm the target user or setting before applying a change.
- Review the result after saving or updating.
- Use the audit view to investigate recorded actions where available.
- Do not share administrator credentials with ordinary users.

10.2 Integration status

The project design includes GitHub, coding-platform, college-LMS and AI-service integrations. The actual availability of each connector depends on environment configuration. The source configuration includes demo connector modes as defaults. An integration status screen should therefore be interpreted as the state of the configured installation, not proof that every external service is connected.

> **Administrative caution**
> Avoid changing integration settings, deleting users or modifying security configuration unless you are authorized and understand the effect. Keep a backup before performing maintenance.

11. Common Problems and Safe Use

Problem	What to check
Cannot sign in	Check the email address and password. Confirm that the account exists and that the authentication service is running.
Registration rejected	Confirm all required fields are filled, the passwords match and the password has at least six characters.
Dashboard does not load	Refresh once, check your network connection and try again. If the issue continues, contact the administrator.
Profile does not save	Check required fields and replace any example values with valid personal information.
Evidence upload fails	Check file type/size requirements shown by the form and confirm that the server is available.
Skills or gap analysis is empty	Confirm that profile/evidence information exists and that the configured skill-analysis service is available.
Learning resources do not appear	Clear filters or search terms and retry. The learning service may be unavailable.
Faculty / recruiter data missing	Confirm the correct role, selected filters and availability of the corresponding backend service.
Unexpected sample values	Check the selected dashboard data mode and whether the installation is using seeded demonstration data.

11.1 Good practices

- Enter accurate information and keep profile details up to date.
- Upload only files you have permission to share.
- Do not upload passwords, identity secrets or unrelated personal information.
- Check verification labels and data mode before relying on scores.
- Sign out when using a shared computer.
- Report incorrect records or suspected security problems to the administrator.

11.2 What to include when reporting a problem

Tell the administrator which page you were using, what action you took, what message appeared, and approximately when it happened. Include a screenshot only if it does not expose passwords, private student data or other sensitive information.

12. Feature Status and Limitations

The project report describes the intended platform, while the supplied source archive shows the implementation snapshot. The following distinctions are important for accurate use of the system.

Area	What is present in the inspected source	User-facing caution
Student registration / login	Login and student registration screens; role-protected routes; demo shortcut in login component.	Demo credentials and access depend on the local configuration and seed data.
Student profile	Profile form and save operation through student service.	Fallback example values exist in the form; verify every field.
Skills and evidence	Skill list, category/search/sort controls, evidence list, add modal, filters and delete/download actions.	Verification depends on available evidence and backend processing.
Skill-gap analysis	Role loading and analysis request through skill service; FastAPI/deterministic source label.	Results depend on the backend and available records.
Learning resources	Resource filtering, start/update progress and completion state.	Tracking reflects information submitted in the application; external completion is not independently guaranteed.
Role matching / progress	Screens are present, but inspected page components include hard-coded illustrative data.	Do not treat their sample percentages or skill rows as live, independently verified scores.
Faculty analytics	Faculty dashboard requests batch analytics from API and provides filters.	Requires a working faculty API and suitable role.
Recruiter matching	Recruiter dashboard loads jobs/candidates and supports shortlist status actions.	Requires configured recruiter API, jobs and consented candidate data.
Admin	Dashboard, user, integration, audit and settings tabs are routed.	Actual available operations depend on backend implementation and permissions.

12.1 Important difference between design and deployment

The project report describes a layered architecture using React/Tailwind, Node.js/Express, MongoDB/Mongoose and Python/FastAPI. The source archive includes frontend, backend and AI-service folders. The presence of a module in the source does not by itself prove that it is deployed, configured, connected to live third-party accounts or validated in a production environment.

12.2 External integrations

The project report lists GitHub, coding platforms, resumes, certificates, academic assessments, projects and internships as potential evidence sources. Connector availability is configuration-dependent; the source example configuration sets GitHub, LeetCode and college LMS connector modes to demo by default. Users should not assume automatic live synchronization unless the administrator confirms it is enabled.

> **Manual version note**
> This manual reflects the supplied project report and ZIP snapshot dated for this preparation (4 November 2026). Update it whenever page names, routes, integrations or user workflows change.

13. Glossary and Project Information

Term	Meaning
AI / ML	Artificial Intelligence / Machine Learning; methods used to analyze available skill information.
Evidence	A supporting item such as a project, certificate, coding profile, assessment or internship record.
Skill gap	A difference between the available skill information and the requirement for a selected role.
Readiness	An indicator of how closely the available skill profile aligns with a role's requirements.
T&P	Training and Placement.
RBAC	Role-Based Access Control; access is limited according to the user's assigned role.
Demo / Sample data	Illustrative records used to demonstrate application pages and workflows.
API	Application Programming Interface; the service through which application parts exchange data.

Project details

Project: SkillBridge – An AI-Powered Skill Intelligence and Placement Platform
Institution: K.K. Wagh Institute of Engineering Education and Research, Nashik
Department: Computer Engineering
Academic Year: 2026–27
Project Group: Group 11
Faculty mentors: Dr. Prof. S. M. Kamalapur and Prof. J. R. Mankar

Project team

Student	Project responsibility (as recorded in report)
Rajas Mahajan	Team Leader and Project Coordinator
Aamrapali More	Testing, Documentation and Deployment
Raj Patil	Backend, AI/NLP and Database Developer
Rajvardhan Patil	Backend, AI/NLP and Database Developer
Soham Patil	UI/UX Designer and Frontend Developer

References and preparation basis

Primary project sources used: (1) SkillBridge PBL Project Report, Group 11, Academic Year 2026–27; and (2) uploaded Skill Bridge source-code ZIP, including frontend route and page components, backend services and configuration examples. The separate ProjectSync report was treated only as a formatting/reference example and was not used as a source of SkillBridge features.

External writing guidance consulted: W3C Web Accessibility Initiative guidance recommends clear instructions, concise language, informative headings and useful text alternatives. Microsoft Style Guide guidance recommends consistent, separately numbered steps for multi-step procedures. These principles informed the manual's presentation.

Guidance links: https://www.w3.org/WAI/tips/writing/ | https://learn.microsoft.com/en-us/style-guide/procedures-instructions/writing-step-by-step-instructions

End of User Manual